InternSync – API Specification

CSC 305 – Software Engineering · Semester 1 – 2026/2027 · Section 5MA1 · Group 3
Task T2 (System design) · Milestone M2 · Version 1.5 · October 28, 2026

Version	Date	Description	Author
1.0	Oct 13, 2026	First version for Milestone M2.	Salam Alsaour
1.1	Oct 20, 2026	CR-05: registration, email confirmation, invitations, verification requests, universities and colleges, company team, Platform Admin endpoints and scope; vacancy review endpoints and CSV import removed.	Salam Alsaour
1.2	Oct 24, 2026	CR-07 (Increment 1 implementation): `GET /admin/universities`, `GET /placements/candidates`, and the development-only demo clock added; endpoints usable before consent listed; registration rate limit; `company_id` filter on placements; audit filters named `date_from` and `date_to`; error codes `STUDENT_NUMBER_TAKEN`, `DUPLICATE_NAME`, `FILE_PURGED`, `SEATS_BELOW_PLACEMENTS`, `ACADEMIC_SUPERVISOR_INVALID`, `COMPANY_SUPERVISOR_INVALID`, and `INTERNAL_ERROR` added; Section 6 describes the permission matrix and the contract check as built.	Salam Alsaour
1.3	Oct 26, 2026	CR-08 (Increment 2 implementation): files are always streamed by the API; `can_*` flags on attendance and grades; reopening a final evaluation needs a `draft` grade; `/version` and the deep health check name the loaded model; dashboard links open the grades page; payloads of Sections 4.5, 4.6, 4.9, and 4.10 as built. Sections 3.7, 3.11, 3.12, 3.19, 3.20, and 4 updated.	Salam Alsaour
1.4	Oct 27, 2026	CR-09 (Increment 3 implementation): transfer rules and answers; similarity result as built (`status` `none`, `sources` with `author`, `can_decide`); notification items with their text in the caller's language; push answers `204`; history entries; exports in the caller's language and audited; error codes `TRANSFER_TARGET_INVALID` and `SUPERVISOR_HAS_STUDENTS`. Sections 3.2, 3.10, 3.15, 3.17, 3.18, 4.8, and 5 updated.	Salam Alsaour
1.5	Oct 28, 2026	CR-11 (Increment 4 implementation): retention and anonymization endpoints; `GET /placements/{id}/feedback` and `/certificate`; the evaluation of the company, survey, outcomes, and leniency answers as built; the public certificate answer with university, college, and `name_removed`; `idle_timeout_minutes` on `/me`; seven new error codes. Sections 3.1, 3.3, 3.12, 3.14, 3.16, 4.11, and 5 updated.	Salam Alsaour

This document is the API contract agreed before implementation (SPMP risk "integration problems"). During development, FastAPI generates the OpenAPI 3.1 document from the code at `/api/v1/openapi.json` (NFR-MNT-02), and the frontend client is generated from it. The generated document must match this specification; any difference is either a bug or a change that is recorded here first.

Related: System Design Document (data model, state machines, security) and SRS 1.5.

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1. Conventions

Topic	Rule
Base URL	`https://<api-host>/api/v1`. Reference deployment: the Hugging Face Space URL; local: `http://localhost:8000/api/v1`.
Transport	HTTPS only (NFR-SEC-01). CORS allows only the configured frontend origins.
Format	JSON in UTF-8. File uploads use `multipart/form-data`. Field names are `snake_case`.
Authentication	`Authorization: Bearer <access_token>` from `POST /auth/login`. The token carries a session ID; the session ends after 30 idle minutes or at logout (FR-AUT-03). Endpoints marked **Public** need no token.
Language	Error messages and generated texts use the user's saved language, or `Accept-Language: ar` / `en` before login. User-entered content is returned as entered (FR-LOC-02).
Identifiers	UUID strings, except `code` of certificates (26-character base32).

Topic	Rule
Dates and times	ISO 8601 with offset. The server returns UTC (`2026-11-05T20:59:00Z`); the client shows Asia/Riyadh time as `DD/MM/YYYY HH:mm` (NFR-L10N-02). Plain dates use `YYYY-MM-DD`.
Enumerations	Lowercase values, the same as the database (for example `revision_requested`, `platform_admin`).
Lists	Paginated with `?page=1&page_size=20` (maximum 100). Response: `{"items": [...], "page": 1, "page_size": 20, "total": 134}`. Sorting with `?sort=field` or `?sort=-field`.
State changes	Each transition of a state machine is its own `POST /<resource>/{id}/<action>` endpoint, so each maps to one row of the transition tables in the design document (Section 5.1). Every successful change writes one audit entry (FR-AUD-01).
Errors	`application/problem+json` (RFC 9457) with a stable `code` (Section 5).
Downloads	Files are never served from a public URL. `GET /files/{id}` checks permission and redirects to a short-lived link (NFR-SEC-04). Export endpoints return the file directly.

Status codes used: `200` OK, `201` created, `202` accepted, `204` no content, `302` redirect to a file, `400` malformed request, `401` not logged in or session ended, `403` role, scope, or consent not allowed, `404` not found, `409` conflict with the current state or a limit, `410` link already used or expired, `413` file too large, `415` file type not allowed, `422` validation failed, `423` account locked, `429` too many requests.

Error body

```
{
  "type": "about:blank",
  "title": "Application limit reached",
  "status": 409,
  "code": "APPLICATION_LIMIT_REACHED",
  "detail": "لديك 5 طلبات نشطة. اسحب طلباً قبل التقديم على فرصة أخرى.",
  "errors": []
}
```

For `422`, `errors` lists each field: `{"field": "weights", "code": "WEIGHTS_NOT_100", "message": "…"}`. Some codes add data, for example `missing` for `PRECONDITIONS_UNMET` (Section 4.9).

2. Roles and Scope Notation

Notation	Meaning
ST, CS, AS, CO, PA	Student, Company Supervisor, Academic Supervisor, Coordinator, Platform Admin
CA	Company Admin: the Company Supervisor who holds the admin right of the company
Public	No login needed
Any	Any logged-in, verified user
(unverified)	Also allowed while the account's verification is pending or rejected
(own)	Only the caller's own records
(own vacancy)	Vacancies whose `company_supervisor_id` is the caller, and their applications
(assigned)	Placements where the caller is the company or academic supervisor, and the records under them
(college)	Records of students of the Coordinator's college, and the college's cycles and users. A plain "CO" below always means this scope
(company)	Records of the Company Admin's company. A plain "CA" below always means this scope
(all)	Every record of its kind on the platform (Platform Admin; never academic records)

A request for a record outside the caller's scope returns `403 FORBIDDEN`. An account that is not yet verified gets `403 VERIFICATION_REQUIRED` from every endpoint not marked (unverified). Each endpoint below is in the permission matrix test (Section 6).

3. Endpoints

3.1 Authentication and Session

Method	Path	Roles	Purpose	SRS
POST	`/auth/register`	Public	Multipart: `kind` (`student`, `coordinator`, `company`), account fields, the fields of that kind, and the documents (Section 4.3). Creates the account and its verification request and emails a confirmation link. At most `auth.registrations_per_ip_hour` registrations per address per hour (`429 RATE_LIMITED`)	FR-ORG-02 – 05, NFR-SEC-03
POST	`/auth/email/confirm`	Public	`{token}`: confirm the email address	FR-AUT-02
POST	`/auth/email/resend`	Public	`{email}`: always `202`; sends a new confirmation link if the account exists and is unconfirmed	FR-AUT-02
POST	`/auth/login`	Public	Log in with email and password; returns the access token and the user. Refused before the email is confirmed (`403 EMAIL_NOT_CONFIRMED`)	FR-AUT-03, NFR-SEC-03
POST	`/auth/logout`	Any (unverified)	End the current session; the token stops working	FR-AUT-03
GET	`/auth/tokens/{token}`	Public	Check a confirmation, invitation, or reset link before showing its form: `{purpose, valid, reason}`	FR-AUT-02, FR-AUT-04
POST	`/auth/invitations/accept`	Public	`{token, password}`: set the password of an invited account	FR-ORG-06
POST	`/auth/password/forgot`	Public	`{email}`: always `202`; sends a reset link if the account exists	FR-AUT-04
POST	`/auth/password/reset`	Public	`{token, password}`: set a new password and end all sessions	FR-AUT-04
GET	`/me`	Any (unverified)	The current user, role, organization, language, `verification_status`, `is_company_admin`, `consent_required`, and `idle_timeout_minutes` (the app applies the same idle limit on the device, CR-11)	FR-AUT-01, FR-AUT-03, FR-ORG-02
PATCH	`/me`	Any (unverified)	`{language}`	FR-LOC-01
POST	`/me/consent`	Any (unverified)	Acknowledge the privacy notice	NFR-PRV-01
GET	`/me/verification`	Any (unverified)	The caller's verification requests with status, reason, and documents	FR-ORG-02, FR-ORG-09
POST	`/me/verification`	Any (unverified)	Multipart: corrected data and documents; a new request after a rejection	FR-ORG-09

Before the privacy notice is acknowledged, only `GET /me`, `PATCH /me` (so the notice can be read in the other language), `POST /me/consent`, and `POST /auth/logout` answer; every other endpoint returns `403 CONSENT_REQUIRED`.

3.2 Organizations, Verification, and Accounts

Method	Path	Roles	Purpose	SRS
GET	`/universities`	Public	Active universities with their active colleges (registration form)	FR-ORG-01
GET	`/admin/universities`	PA	Every university with every college, active or not (administration page)	FR-ORG-01
POST	`/universities`	PA	Add a university: names, email domains, GPA scale	FR-ORG-01

Method	Path	Roles	Purpose	SRS
PATCH	`/universities/{id}`	PA	Edit or deactivate	FR-ORG-01
POST	`/universities/{id}/colleges`	PA	Add a college	FR-ORG-01
GET	`/colleges/{id}`	PA, CO	College with its Coordinator and counts	FR-ORG-01
PATCH	`/colleges/{id}`	PA	Edit or deactivate	FR-ORG-01
POST	`/colleges/{id}/coordinator/transfer`	PA	`{user_id, reason}`: the new coordinator must be an active academic supervisor of the college who has set a password (`422 TRANSFER_TARGET_INVALID`) and has no active students (`409 SUPERVISOR_HAS_STUDENTS`); the previous coordinator becomes an academic supervisor. Answers the college	FR-ORG-07
GET	`/colleges/{id}/academic-supervisors`	CO, PA	The college's academic supervisors with student counts	FR-ORG-06
POST	`/colleges/{id}/invitations`	CO	`{email, name_en, name_ar}`: invite an academic supervisor	FR-ORG-06
GET	`/companies`	PA (all), CS (own)	List companies: `?status&q`	FR-ORG-04
GET	`/companies/{id}`	PA, CS (own)	Company profile with status	FR-ORG-04
PATCH	`/companies/{id}`	CA	City and website	FR-ORG-04
POST	`/companies/{id}/suspend`	PA	`{reason}`: closes its published vacancies	FR-AUT-06
POST	`/companies/{id}/reactivate`	PA	Reactivate	FR-AUT-06
POST	`/companies/{id}/admin/transfer`	PA	`{user_id, reason}`: the new Company Admin must be an active supervisor of the company who has set a password (`422 TRANSFER_TARGET_INVALID`). Answers the company's supervisors	FR-ORG-07
GET	`/companies/{id}/supervisors`	CA, PA	The company's supervisors with trainee counts	FR-ORG-06
POST	`/companies/{id}/invitations`	CA	`{email, name_en, name_ar}`: invite a company supervisor	FR-ORG-06
GET	`/verification-requests`	PA (coordinator and company requests), CO (student requests of the college)	`?status&kind`	FR-ORG-03 – 05, FR-DSH-01
GET	`/verification-requests/{id}`	The reviewer; the applicant	Submitted data, earlier decisions of the same account, document links	FR-ORG-09

Method	Path	Roles	Purpose	SRS
POST	`/verification-requests/{id}/approve`	PA (coordinator, company), CO (student)	Student requests: `{cycle_id}` of an open cycle of the college	FR-ORG-03 – 05
POST	`/verification-requests/{id}/reject`	PA (coordinator, company), CO (student)	`{reason}`	FR-ORG-03 – 05
GET	`/users`	PA (account fields), CO, CA	Search accounts: `?role&q&is_active&verification_status`	FR-AUT-06
GET	`/users/{id}`	PA, CO, CA	Account details	FR-AUT-06
POST	`/users/{id}/deactivate`	PA (any), CO (the college's academic supervisors and students), CA (the company's supervisors)	Deactivate and end all sessions	FR-AUT-06
POST	`/users/{id}/reactivate`	Same	Reactivate	FR-AUT-06
POST	`/users/{id}/invitation/resend`	CO, CA	New invitation link for an account that has not accepted it; the old one is revoked	FR-ORG-06

3.3 Training Cycles

Method	Path	Roles	Purpose	SRS
GET	`/cycles`	Any (ST: own enrolment; CS, AS: cycles of their records; CO: the college's)	List cycles	FR-CYC-01
POST	`/cycles`	CO	Create a cycle for the college; generates its weeks and deadlines	FR-CYC-01
GET	`/cycles/{id}`	Same as list	Cycle details	FR-CYC-01
PATCH	`/cycles/{id}`	CO	Edit; dates, weeks, working days, and report day and time only before the first placement (`409 SCHEDULE_LOCKED`)	FR-CYC-01
GET	`/cycles/{id}/weeks`	Same as list	Weeks with dates, report deadline, scheduled days	FR-CYC-01
PATCH	`/cycles/{id}/weeks/{week_no}`	CO	`{report_due_at, reason}`: move one week's deadline; updates the slots still in draft	FR-CYC-01
POST	`/cycles/{id}/close`	CO	Make the cycle read-only	FR-CYC-05
POST	`/cycles/{id}/reopen`	CO	`{reason}`; refused after an anonymization (`409 ALREADY_ANONYMIZED`)	FR-CYC-05
GET	`/cycles/{id}/anonymization`	CO	`status` (`open`, `waiting`, `ready`, `done`), `closed_at`, `eligible_at`, `retention_months`, the number of `students` to anonymize and of `skipped` ones, `anonymized_at`	NFR-PRV-04

Method	Path	Roles	Purpose	SRS
POST	`/cycles/{id}/anonymize`	CO	`{confirm: true}` (`422 CONFIRM_REQUIRED` without it); `409 CYCLE_NOT_CLOSED`, `409 RETENTION_NOT_OVER` with `eligible_at`, `409 ALREADY_ANONYMIZED` when nobody is left; answers as the `GET`	NFR-PRV-04
GET	`/colleges/{id}/retention`	CO (own college)	`{retention_months}`	NFR-PRV-04
PUT	`/colleges/{id}/retention`	CO (own college)	`{retention_months}`, 0–120; audited	NFR-PRV-04
GET	`/cycles/{id}/grade-components`	CO, AS	Components and weights	FR-CYC-02
PUT	`/cycles/{id}/grade-components`	CO	Replace the list; total must be 100 (`422 WEIGHTS_NOT_100`); refused after a grade is submitted (`409 GRADES_EXIST`)	FR-CYC-02
GET	`/cycles/{id}/rubric`	Any in the cycle	Criteria with the five level descriptions	FR-CYC-03
PUT	`/cycles/{id}/rubric`	CO	Replace the criteria; refused after the first evaluation (`409 RUBRIC_LOCKED`)	FR-CYC-03
GET	`/cycles/{id}/enrollments`	CO	Enrolled students (created by approving their verification)	FR-ORG-05

3.4 Skill Vocabulary, Majors, and Settings

Method	Path	Roles	Purpose	SRS
GET	`/skills`	Any (unverified)	Search skills: `?q` matches both names and synonyms	FR-CYC-04
POST	`/skills`	PA	`{name_en, name_ar, synonyms}`	FR-CYC-04
PATCH	`/skills/{id}`	PA	Edit names, synonyms, `is_active`	FR-CYC-04
GET	`/skills/{id}/resources`	Any (unverified)	Learning resources	FR-CYC-04
PUT	`/skills/{id}/resources`	PA	Replace the resources (title, URL, effort, language)	FR-CYC-04
GET	`/majors`	Public	List majors (registration form)	FR-PRF-02, FR-VAC-01
POST	`/majors`	PA	Add a major	FR-VAC-01
PATCH	`/majors/{id}`	PA	Edit names	FR-VAC-01
GET	`/settings`	PA	All configurable defaults with their values	NFR-MNT-03
PATCH	`/settings`	PA	`{ "<key>": <value>, … }`; each value is validated; takes effect at once	NFR-MNT-03

3.5 Student Profile and CV

Method	Path	Roles	Purpose	SRS
GET	`/students/me/profile`	ST (unverified)	Profile: student number, major, GPA, and level (read-only after verification), languages, interests, confirmed skills, current CV	FR-PRF-02
PATCH	`/students/me/profile`	ST (unverified)	`{languages, interests}`; student number, major, and GPA are refused after verification (`422 READ_ONLY_FIELD`)	FR-PRF-02
GET	`/students/me/cvs`	ST (unverified)	CV history	FR-PRF-03
POST	`/students/me/cvs`	ST (unverified)	Multipart `file` (PDF or DOCX, ≤ 5 MB): stores the CV, extracts text and skills	FR-PRF-03 – 05, NFR-PER-03

Method	Path	Roles	Purpose	SRS
GET	`/students/me/skills`	ST (unverified)	Confirmed skills	FR-PRF-04
PUT	`/students/me/skills`	ST (unverified)	`{skills: [{skill_id, source}]}`: confirm the list; recomputes the profile embedding	FR-PRF-04
GET	`/students/{id}/profile`	CS (applicant to own vacancy, or assigned), CA, AS (assigned), CO	Profile, confirmed skills, current CV file	FR-VAC-06

3.6 Vacancies

Method	Path	Roles	Purpose	SRS
GET	`/vacancies`	ST (unverified; published and open to the student's university), CS (own company), CO (published, read-only), PA (all, read-only)	Browse: `?q&city&company_id&major_id&skill_id&training_type&status`. For ST each item carries `eligible` and `ineligible_reason`	FR-VAC-03, FR-MAT-03
POST	`/vacancies`	CS of a verified company	Create a vacancy as `draft`, or published at once with `publish: true`; `target_university_ids` optional	FR-VAC-01, FR-VAC-02
GET	`/vacancies/{id}`	Same as list	Details with required and preferred skills, eligible majors, and target universities	FR-VAC-01
PATCH	`/vacancies/{id}`	CS (own vacancy), CA	Edit; a published vacancy is embedded again; seats cannot drop below the confirmed placements	FR-VAC-01
POST	`/vacancies/{id}/publish`	CS (own vacancy), CA	`draft` → `published`	FR-VAC-02
POST	`/vacancies/{id}/close`	CS (own vacancy), CA	`published` → `closed`	FR-VAC-02

3.7 Applications and Placements

Method	Path	Roles	Purpose	SRS
POST	`/vacancies/{id}/applications`	ST	Apply; stores the fit-score snapshot	FR-VAC-04
GET	`/vacancies/{id}/applications`	CS (own vacancy), CA	Applicants with university, college, profile summary, confirmed skills, fit snapshot (`fit_score`, `match_mode`)	FR-VAC-06
GET	`/applications`	ST (own), CO (students of the college)	`?status&cycle_id&placed=false` (CO: accepted applicants waiting for a placement)	FR-VAC-04, FR-VAC-07
GET	`/applications/{id}`	ST (own), CS (own vacancy), CA, CO	One application	FR-VAC-06
POST	`/applications/{id}/withdraw`	ST (own)	`submitted` → `withdrawn`	FR-VAC-05
POST	`/applications/{id}/accept`	CS (own vacancy), CA	`submitted` → `accepted`; notifies the student's Coordinator	FR-VAC-06

Method	Path	Roles	Purpose	SRS
POST	`/applications/{id}/decline`	CS (own vacancy), CA	`submitted` → `declined`	FR-VAC-06
GET	`/placements/candidates`	CO	Accepted applicants of the college waiting for a placement, each with the vacancy's supervisor (the default company supervisor) and the company's other supervisors	FR-VAC-07
POST	`/placements`	CO (the student's college)	`{application_id, academic_supervisor_id, company_supervisor_id?}`: confirm in the student's cycle; creates report slots and the draft grade	FR-VAC-07, FR-REP-01
GET	`/placements`	ST (own), CS (assigned), CA, AS (assigned), CO	`?cycle_id&company_id&academic_supervisor_id&status`; each item with report counts, `attendance_percent`, and `grade_status`	FR-VAC-07
GET	`/placements/{id}`	Same as list	Overview: student, vacancy, supervisors, status, attendance %, report summary, grade status	FR-VAC-07, FR-ATT-03
PATCH	`/placements/{id}/academic-supervisor`	CO	`{academic_supervisor_id, reason}`	FR-VAC-08
POST	`/placements/{id}/cancel`	CO	`{reason}`	FR-VAC-08

3.8 Matching

Method	Path	Roles	Purpose	SRS
GET	`/students/me/matches`	ST	Ranked eligible vacancies: `?limit`. Needs a verified, enrolled student (`403 VERIFICATION_REQUIRED`) and confirmed skills (`409 SKILLS_NOT_CONFIRMED`)	FR-MAT-01 – 05, FR-MAT-07, NFR-PER-02
GET	`/students/me/matches/{vacancy_id}`	ST	One vacancy's fit with its explanation	FR-MAT-05
GET	`/students/me/matches/{vacancy_id}/plan`	ST	Development plan for the missing skills	FR-MAT-06

3.9 Weekly and Final Reports

Method	Path	Roles	Purpose	SRS
GET	`/placements/{id}/report-slots`	ST (own), CS (assigned, read), AS (assigned), CO	All slots with status, deadline, Late and Missing flags	FR-REP-01, FR-REP-04
GET	`/report-slots/{id}`	Same	Slot with submitted versions, reviews, and attachments; the draft only for ST	FR-REP-03, FR-REP-05
PUT	`/report-slots/{id}/draft`	ST (own)	Create or update the draft's text fields	FR-REP-03
POST	`/report-slots/{id}/draft/attachments`	ST (own)	Multipart `file` and `role` (`main` or `attachment`)	FR-REP-02
DELETE	`/report-slots/{id}/draft/attachments/{attachment_id}`	ST (own)	Remove an attachment from the draft	FR-REP-02
POST	`/report-slots/{id}/submit`	ST (own)	Submit or resubmit the draft; records server time and the Late flag; starts the similarity check	FR-REP-02, FR-REP-04, FR-SIM-01
POST	`/report-slots/{id}/accept`	AS (assigned)	`submitted` → `accepted`	FR-REP-05
POST	`/report-slots/{id}/request-revision`	AS (assigned)	`{comment}`: `submitted` → `revision_requested`	FR-REP-05
POST	`/report-slots/{id}/close-missing`	AS (assigned)	`draft` → `closed_missing` after the deadline	FR-REP-07

Method	Path	Roles	Purpose	SRS
GET	`/report-slots/{id}/comments`	ST (own), CS (assigned, read), AS (assigned), CO	Comments with author and time	FR-REP-06
POST	`/report-slots/{id}/comments`	ST (own), AS (assigned)	`{body}`	FR-REP-06
GET	`/report-versions/{id}`	ST (own), CS, AS (assigned), CO	One version; drafts only for ST	FR-REP-05

3.10 Similarity

Method	Path	Roles	Purpose	SRS
GET	`/report-versions/{id}/similarity`	AS (assigned), CO	Status (`none`, `pending`, `done`, `failed`), score, flag, model version, sources with their passage pairs; the source author only for the coordinator of the author's college; `can_decide`	FR-SIM-02
POST	`/report-versions/{id}/similarity/decision`	AS (assigned)	`{decision: no_issue | overlap_confirmed, comment}`; only for flagged checks	FR-SIM-03
GET	`/similarity-checks`	AS (assigned), CO	`?cycle_id&flagged=true&decided=false`	FR-DSH-01

3.11 Attendance

Method	Path	Roles	Purpose	SRS
GET	`/placements/{id}/attendance`	ST (own), CS, AS (assigned), CO	Weeks with recorded days, state (`upcoming`, `open`, `pending`, `overdue`, `recorded`) and `locked`, the percentage, and `can_record` and `can_unlock` for the caller	FR-ATT-03, FR-ATT-04
PUT	`/placements/{id}/attendance/{week_no}`	CS (assigned)	`{attended_days, note}` or `{full: true}`; refused before the week starts (`409 WEEK_NOT_STARTED`), when locked (`409 WEEK_LOCKED`), or with more days than scheduled (`422 ATTENDED_EXCEEDS_SCHEDULED`)	FR-ATT-01, FR-ATT-02
POST	`/placements/{id}/attendance/{week_no}/unlock`	CO	`{reason, until}` (default: 7 days from now)	FR-ATT-02

3.12 Evaluations

Method	Path	Roles	Purpose	SRS
GET	`/placements/{id}/evaluations`	CS, AS (assigned), CO; ST (own: periodic when submitted, final only when the grade is Final)	Evaluations with scores	FR-EVL-03
PUT	`/placements/{id}/evaluations/{type}`	CS (assigned)	Save a draft: `{scores: [{criterion_id, score}], overall_comment}`; `type` is `periodic` or `final`	FR-EVL-01
POST	`/placements/{id}/evaluations/{type}/submit`	CS (assigned)	Submit; every criterion and the comment are required; locks the rubric	FR-EVL-01, FR-CYC-03
POST	`/evaluations/{id}/reopen`	CO	`{reason}`; once per evaluation (`409 ALREADY_REOPENED`); a final evaluation only while its grade is `draft` (`409 INVALID_TRANSITION`)	FR-EVL-04
GET	`/placements/{id}/feedback`	ST (own), CO	Both answers below, the company and vacancy, `open_from`, `is_open`, `can_evaluate`, and `can_answer`	FR-EVL-05, FR-DSH-03

Method	Path	Roles	Purpose	SRS
GET	`/placements/{id}/company-evaluation`	ST (own), CO	The student's evaluation of the training entity, or `null`	FR-EVL-05
PUT	`/placements/{id}/company-evaluation`	ST (own)	The four ratings 1–5, `would_recommend`, optional `comment`; once (`409 ALREADY_SUBMITTED`), from the training end date or the Final grade (`409 FEEDBACK_NOT_OPEN`); answers as the feedback `GET`	FR-EVL-05
GET	`/cycles/{id}/company-evaluations`	CO	Per company and for the cycle: answers, the mean of each rating, and `recommend_rate`	FR-EVL-05
GET	`/placements/{id}/outcome-survey`	ST (own), CO	Job-offer answer, or `null`	FR-DSH-03
PUT	`/placements/{id}/outcome-survey`	ST (own)	`{job_offer: yes | no | pending}`; can be changed later; answers as the feedback `GET`	FR-DSH-03

3.13 Grades

Method	Path	Roles	Purpose	SRS
GET	`/placements/{id}/grade`	AS (assigned), CO; ST (own, only when Final, otherwise `{"visible": false}`)	Components, total, status, history link	FR-WFL-02, FR-WFL-06
PUT	`/placements/{id}/grade`	AS (assigned)	`{component_scores: [{component_id, score}]}` while `draft`; the company component is computed	FR-WFL-02
GET	`/placements/{id}/grade/preconditions`	AS (assigned), CO	Checklist of the FR-WFL-03 conditions	FR-WFL-03
POST	`/grades/{id}/submit`	AS (assigned)	`draft` → `submitted`, or `409 PRECONDITIONS_UNMET` with the missing items	FR-WFL-03
POST	`/grades/{id}/approve`	CO	`submitted` → `final`	FR-WFL-04
POST	`/grades/{id}/return`	CO	`{comment}`: `submitted` → `draft`	FR-WFL-04
POST	`/grades/{id}/reopen`	CO	`{reason}`: `final` → `draft`	FR-WFL-05
POST	`/grades/bulk-approve`	CO	`{grade_ids}`; one audit entry per grade; result per grade	FR-WFL-07
GET	`/grades`	AS (assigned), CO	`?cycle_id&status`	FR-DSH-01

3.14 Dashboards and Statistics

Method	Path	Roles	Purpose	SRS
GET	`/dashboard`	Any	Pending actions of the caller's role with counts and links	FR-DSH-01
GET	`/cycles/{id}/stats`	CO	Cycle status overall and per company: `?company_id`	FR-DSH-02
GET	`/cycles/{id}/outcomes`	CO	Placement-to-job-offer rate per cycle and per company: `answers`, `yes`, `no`, `pending`, `job_offer_rate` (Yes ÷ answers)	FR-DSH-03
GET	`/cycles/{id}/leniency-alerts`	CO	`cycle_mean`, and per company its final evaluations, `mean`, `difference`, and `alert` (at least `dashboard.leniency_min_evaluations` and a difference of at least `dashboard.leniency_delta`)	FR-DSH-04
GET	`/escalations`	CO	Items overdue 3 days or more: `?cycle_id&type`	FR-NOT-04

Method	Path	Roles	Purpose	SRS
GET	`/platform/stats`	PA	Universities, colleges with and without a coordinator, verified companies, users by role, pending verification requests and their age, published vacancies; no academic records	FR-DSH-05

3.15 Exports

Method	Path	Roles	Purpose	SRS
GET	`/cycles/{id}/exports/grade-sheet?format=xlsx|pdf`	CO	Grade sheet file, as an attachment in the caller's language; every export is audited (`export`)	FR-EXP-01, NFR-PER-05
GET	`/cycles/{id}/exports/term-summary`	CO	Term summary PDF	FR-EXP-02
GET	`/placements/{id}/exports/training-record`	CO, AS (assigned)	Student training record PDF	FR-EXP-03

3.16 Certificates

Method	Path	Roles	Purpose	SRS
GET	`/placements/{id}/certificate`	CO, ST (own)	`{certificate, can_issue}`: the valid certificate, else the latest revoked one, or `null`	FR-CER-01
POST	`/placements/{id}/certificate`	CO	Issue a certificate; the grade must be Final; `201` with the same answer as the `GET`	FR-CER-01
GET	`/certificates`	CO	`?cycle_id&status`	FR-CER-01
GET	`/certificates/{id}`	CO, ST (own)	Certificate details and file	FR-CER-01
POST	`/certificates/{id}/revoke`	CO	`{reason}`	FR-CER-03
GET	`/public/certificates/{code}`	Public	Status (`valid`, `revoked`, `not_found`), student name, university and college, cycle, company, training dates; never the grade	FR-CER-02

3.17 Notifications and Push

Method	Path	Roles	Purpose	SRS
GET	`/notifications`	Any (own)	`?unread_only&page&page_size`; each item has `title` and `body` already in the caller's language (FR-NOT-06), `link`, `read_at`; includes `unread_count`	FR-NOT-01
POST	`/notifications/{id}/read`	Any (own)	Mark one as read	FR-NOT-01
POST	`/notifications/read-all`	Any (own)	Mark all as read	FR-NOT-01
GET	`/push/vapid-public-key`	Any	`{key}`: the public key for the browser subscription, `null` when push is not configured	FR-PWA-02
POST	`/push/subscriptions`	Any (own)	`{endpoint, keys: {p256dh, auth}}`; `204`. The same endpoint signed in as another user moves to that user	FR-PWA-02
DELETE	`/push/subscriptions`	Any (own)	`{endpoint}`; `204`	FR-PWA-02

3.18 Audit and History

Method	Path	Roles	Purpose	SRS
GET	`/audit-log`	CO (the college's entries), PA (entries about accounts, organizations, and verification)	`?student_id&user_id&entity_type&entity_id&date_from&date_to`; read-only, no update or delete endpoint exists	FR-AUD-02

Method	Path	Roles	Purpose	SRS
GET	`/history/{entity_type}/{entity_id}`	Anyone allowed to view the record; `entity_type` is `report_slot`, `evaluation`, or `grade`	Entries oldest first: `{at, action, role, user_en, user_ar, old, new, reason}`. The student sees a grade's history only when it is Final and never the similarity decision	FR-AUD-03

3.19 Files

Method	Path	Roles	Purpose	SRS
GET	`/files/{id}`	Anyone allowed to view the record that owns the file; for verification documents only the uploader and the reviewer	The file itself, streamed by the API from either storage with `Cache-Control: private, no-store`; no link to the bucket is ever given out. `403` otherwise; `410` for a purged verification document	NFR-SEC-04, FR-ORG-08

3.20 System

Method	Path	Roles	Purpose	SRS
GET	`/health`	Public	`{status}`; with `?deep=true` also `database`, `model` (the loaded model version, `null` while it loads or when it is off), and `scheduler_last_run` (keep-alive and uptime log)	NFR-REL-01
GET	`/version`	Public	`{version, embedding_model}`: the application version and the loaded model version (`null` while it loads or when it is off)	NFR-AI-02
GET	`/openapi.json`, `/docs`	Public	Generated OpenAPI document and its viewer	NFR-MNT-02
GET	`/dev/clock`	Public, development only	The server's current time and the demo offset	–
POST	`/dev/clock/advance`	Public, development only	`{hours}`: move the server clock forward for the M3 demo (late and missing reports)	–
POST	`/dev/clock/reset`	Public, development only	Back to real time	–

The `/dev` routes are mounted only when `ENV` is `development` or `test`; the deployed API and the committed OpenAPI document do not contain them.

4. Main Payloads

Examples show the main fields; the generated OpenAPI document holds the complete schemas.

4.1 Login

`POST /auth/login`

```
{ "email": "sara.ahmed@example.edu", "password": "••••••••" }
```

`200 OK`

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIs…",
  "token_type": "bearer",
  "expires_at": "2026-11-05T08:00:00Z",
  "idle_timeout_minutes": 30,
  "user": {
    "id": "6f1c…", "role": "student", "name_en": "Sara Ahmed", "name_ar": "سارة أحمد",
```

```
    "language": "ar", "consent_required": true,
    "verification_status": "verified", "is_company_admin": false,
    "organization": { "university": "Example University", "college": "College of Computer Science", "college_id": "3c1d…"
}
  }
}
```

Errors: `401 INVALID_CREDENTIALS`, `423 ACCOUNT_LOCKED` (with `locked_until`), `403 ACCOUNT_INACTIVE`, `403 EMAIL_NOT_CONFIRMED`.

4.2 Create a Training Cycle

`POST /cycles`

```
{
  "name": "Co-op Fall 2026", "type": "coop",
  "start_date": "2026-09-06", "end_date": "2026-11-28", "weeks": 12,
  "working_days": [7, 1, 2, 3, 4],
  "report_weekday": 4, "report_time": "23:59",
  "periodic_eval_due_at": "2026-10-15T20:59:00Z",
  "final_eval_due_at": "2026-11-26T20:59:00Z",
  "final_report_due_at": "2026-11-26T20:59:00Z"
}
```

`201 Created` returns the cycle with its `weeks`: `[{"week_no": 1, "starts_on": "2026-09-06", "ends_on": "2026-09-12", "report_due_at": "2026-09-10T20:59:00Z", "scheduled_days": 5}, …]`.

4.3 Registration and Verification

`POST /auth/register` (multipart) for each kind:

Kind	Fields	Documents
`student`	`email`, `password`, `name_en`, `name_ar`, `college_id`, `student_number`, `major_id`, `gpa`, `level`	`course_registration` (required)
`coordinator`	`email`, `password`, `name_en`, `name_ar`, `college_id`	`appointment_letter` (required)
`company`	`email`, `password`, `name_en`, `name_ar`, `company_name`, `commercial_reg_no`, `city`, `website`	`commercial_registration` and `authorization_letter` (both required)

`201 Created`

```
{
  "user_id": "a41e…", "kind": "student", "email_confirmation_sent": true,
  "verification": { "request_id": "77c0…", "status": "pending",
                    "reviewer": "coordinator", "college": "College of Computer Science" }
}
```

Errors: `409 EMAIL_TAKEN`, `409 COLLEGE_HAS_COORDINATOR`, `409 DUPLICATE_COMMERCIAL_REG`, `422 DOCUMENT_REQUIRED`, `422 GPA_ABOVE_SCALE`, `429 RATE_LIMITED`.

`GET /verification-requests?status=pending` (as a Coordinator) → items such as:

```
{
  "id": "77c0…", "kind": "student", "status": "pending", "submitted_at": "2026-10-19T08:40:00Z",
  "applicant": { "name_en": "Sara Ahmed", "name_ar": "سارة أحمد", "email": "sara.ahmed@example.edu",
                 "email_matches_university_domain": true },
  "submitted_data": { "student_number": "4400012345", "major": "Computer Science", "gpa": 4.21, "level": 8 },
  "documents": [ { "kind": "course_registration", "file_id": "f9…", "original_name": "schedule.pdf" } ],
  "earlier_decisions": []
}
```

`POST /verification-requests/{id}/approve` with `{"cycle_id": "0a9d…"}` → `200 OK` with the request `approved` and the new enrolment; `409 CYCLE_NOT_OPEN` if the cycle is closed. `POST /verification-requests/{id}/reject` with `{"reason": "…"}` → `200 OK`; `422 REASON_REQUIRED` without a reason.

4.4 CV Upload

`POST /students/me/cvs` (multipart `file`) → `201 Created`

```
{
  "cv": { "id": "9c3a…", "version_no": 2, "file_id": "e41b…", "uploaded_at": "2026-10-20T09:12:44Z" },
  "text_status": "ok",
  "language": "mixed",
  "extracted_skills": [
    { "skill_id": "1d2e…", "name_en": "JavaScript", "name_ar": "جافا سكربت", "evidence": "…built a dashboard in JS and
React…" },
    { "skill_id": "7a8b…", "name_en": "SQL", "name_ar": "إس كيو إل", "evidence": "…قواعد بيانات SQL…" }
  ],
  "advisory": true
}
```

With `text_status: "no_text"` the list is empty and the client shows the FR-PRF-05 message with manual entry.

4.5 Ranked Vacancies

`GET /students/me/matches?limit=20`

```
{
  "mode": "semantic",
  "model_version": "intfloat/multilingual-e5-small@<revision>",
  "advisory": true,
  "total": 1,
  "items": [
    {
      "vacancy": { "id": "51f0…", "title": "Junior Data Analyst", "company": "Example Co.", "city": "Dammam",
                   "application_deadline": "2026-09-01T20:59:00Z" },
      "fit_score": 82.40,
      "mode": "semantic",
      "parts": { "semantic": 0.78, "required": 1.0, "preferred": 0.5 },
      "matched_skills": [ { "id": "7a8b…", "name_en": "SQL", "name_ar": "إس كيو إل" } ],
      "missing_skills": [],
      "already_applied": false
    }
  ]
}
```

In basic mode `mode` is `"basic"`, `parts.semantic` is `null`, and the client shows the "Basic matching" label (FR-MAT-04). Each item has its own `mode`: a vacancy without an embedding is scored in basic mode, and the list's `mode` is then `"mixed"`. `parts.preferred` is `null` when the vacancy lists no preferred skill; its weight then moves to the required part. `matched_skills` and `missing_skills` compare the confirmed skills with the vacancy's required skills (FR-MAT-05).

4.6 Development Plan

`GET /students/me/matches/{vacancy_id}/plan`

```
{
  "vacancy_id": "51f0…",
  "missing_skills": [
    { "skill": { "id": "c0d1…", "name_en": "Power BI", "name_ar": "باور بي آي" },
      "resources": [ { "title": "Power BI for beginners", "url": "https://…", "estimated_effort": "8 hours", "language":
"en" } ] },
    { "skill": { "id": "f9e8…", "name_en": "Statistics", "name_ar": "الإحصاء" },
      "resources": [] }
  ],
  "advisory": true
}
```

An empty `resources` list is shown by the client as "No resource available yet" in the user's language.

4.7 Report Slot

`GET /report-slots/{id}` (as the academic supervisor)

```
{
  "id": "3e5f…", "placement_id": "a1b2…", "kind": "weekly", "week_no": 3,
  "due_at": "2026-09-24T20:59:00Z", "status": "submitted",
  "is_late": true, "is_missing": true,
  "versions": [
    { "id": "77aa…", "version_no": 1, "submitted_at": "2026-09-25T06:30:12Z", "is_late": true,
      "tasks_performed": "…", "learning": "…", "challenges": "…", "next_week_plan": "…",
      "attachments": [ { "id": "c1…", "file_id": "f1…", "original_name": "diagram.png", "role": "attachment" } ],
      "similarity": { "status": "done", "max_score": 0.12, "is_flagged": false },
      "review": null }
  ],
  "allowed_actions": ["accept", "request_revision"]
}
```

`allowed_actions` lists the transitions the caller may run now, so the client shows only valid buttons.

4.8 Similarity Result

`GET /report-versions/{id}/similarity` (as the academic supervisor)

```
{
  "status": "done", "max_score": 0.46, "is_flagged": true, "threshold": 0.30,
  "model_version": "intfloat/multilingual-e5-small@<revision>",
  "checked_at": "2026-10-27T00:28:41Z", "advisory": true, "can_decide": true,
  "sources": [
    { "version_id": "5d6c…", "slot_id": "8e1a…", "kind": "weekly", "week_no": 3, "version_no": 1,
      "cycle_id": "c0a2…", "cycle_name": "Co-op Spring 2026", "score": 0.46, "author": null,
      "passages": [ { "report": [120, 214], "report_text": "…", "source": [88, 182], "source_text": "…", "cosine": 0.97 }
] }
  ],
  "decision": null, "decision_comment": null
}
```

For the coordinator of the source author's college, `author` is `{"id": "…", "name_en": "…", "name_ar": "…", "student_number": "…"}` (FR-SIM-02). Before a check exists `status` is `none`; while it runs, `pending`; without the model, `failed` until the retry job runs it.

4.9 Grade Submission Refused

`POST /grades/{id}/submit` → `409 Conflict`

```
{
  "status": 409, "code": "PRECONDITIONS_UNMET",
  "title": "The grade cannot be submitted yet. Complete the items listed first.",
  "detail": "لا يمكن إرسال الدرجة بعد. أكمل العناصر المذكورة أولاً.",
  "errors": [],
  "missing": [
    { "item": "final_evaluation", "week_no": null, "component_id": null },
    { "item": "weekly_report", "week_no": 7, "component_id": null },
    { "item": "component", "week_no": null, "component_id": "3c4d…" }
  ]
}
```

`item` is `final_evaluation`, `final_report`, `weekly_report` (with `week_no`), or `component` (an academic component without a score, with `component_id`). The client words each item in the user's language; `GET /placements/{id}/grade/preconditions` returns the same items with `ok`.

4.10 Dashboard

`GET /dashboard` (as the academic supervisor)

```
{
  "role": "academic_supervisor",
  "items": [
    { "type": "reports_to_review", "count": 6, "link": "/academic/students" },
    { "type": "grades_to_submit", "count": 2, "link": "/academic/grades?status=draft" },
    { "type": "grades_returned", "count": 0, "link": "/academic/grades?status=draft" }
  ]
}
```

The item types of each role are listed in the Increment 2 design, Section 8; later increments add theirs (for example `open_similarity_flags` with similarity detection). An item may carry extra fields: `status` on the student's `verification` item, `oldest_days` on the Platform Admin's `verifications_pending`.

4.11 Public Certificate Verification

`GET /public/certificates/7K2M9QX4…` → `200 OK`

```
{ "status": "valid", "name_removed": false,
  "student_name": { "en": "Sara Ahmed", "ar": "سارة أحمد" },
  "university": { "en": "Imam Abdulrahman Bin Faisal University", "ar": "جامعة الإمام عبدالرحمن بن فيصل" },
  "college": { "en": "College of Computer Science and Information Technology", "ar": "كلية علوم الحاسب وتقنية المعلومات" },
  "cycle": "Co-op Fall 2026", "company": "Example Co.",
  "training_start": "2026-09-06", "training_end": "2026-11-28" }
```

An unknown or altered code returns `200` with `{"status": "not_found"}` and the other fields `null`, so the page can always show a clear result (FR-CER-02). The code is accepted with or without dashes and in any case. After the student's data was anonymized (NFR-PRV-04), `student_name` is `null` and `name_removed` is `true`; the status still answers.

5. Error Codes

Code	HTTP	When	SRS
`VALIDATION_ERROR`	422	Field-level validation failed (see `errors`)	NFR-USA-05
`UNAUTHENTICATED`	401	Missing or invalid token	FR-AUT-03
`SESSION_EXPIRED`	401	Idle timeout, logout, or maximum session length reached	FR-AUT-03
`FORBIDDEN`	403	Role or record scope not allowed	FR-AUT-05
`CONSENT_REQUIRED`	403	Privacy notice not yet acknowledged	NFR-PRV-01
`NOT_FOUND`	404	Record does not exist	–
`INVALID_TRANSITION`	409	Action not allowed from the current state	FR-WFL-01
`CYCLE_CLOSED`	409	Write to a closed cycle	FR-CYC-05
`RATE_LIMITED`	429	Too many registrations from one address in an hour	NFR-SEC-03
`INVALID_CREDENTIALS`	401	Wrong email or password	FR-AUT-03
`ACCOUNT_LOCKED`	423	5 failed logins; includes `locked_until`	NFR-SEC-03
`ACCOUNT_INACTIVE`	403	Deactivated account	FR-AUT-06
`EMAIL_NOT_CONFIRMED`	403	Login before the email address is confirmed	FR-AUT-02
`VERIFICATION_REQUIRED`	403	The account's verification is pending or rejected, or the student is not enrolled	FR-ORG-02
`EMAIL_TAKEN`	409	Registration or invitation with an email already in use	FR-ORG-02
`STUDENT_NUMBER_TAKEN`	409	Student number already used at the same university	FR-ORG-05
`DUPLICATE_NAME`	409	A university, college, skill, major, or cycle with the same name already exists	FR-ORG-01, FR-CYC-01, FR-CYC-04
`COLLEGE_HAS_COORDINATOR`	409	Coordinator registration for a college that already has one	FR-ORG-03
`DUPLICATE_COMMERCIAL_REG`	409	Company registration with a registration number already on the platform	FR-ORG-04
`DOCUMENT_REQUIRED`	422	A required verification document is missing	FR-ORG-03 – 05
`GPA_ABOVE_SCALE`	422	GPA above the university's scale	FR-ORG-05
`CYCLE_NOT_OPEN`	409	Student approval with a closed cycle	FR-ORG-05

Code	HTTP	When	SRS
`COMPANY_NOT_VERIFIED`	409	Vacancy or invitation by a company that is not verified, or is suspended	FR-ORG-04, FR-VAC-01
`TOKEN_INVALID`	400	Unknown confirmation, invitation, or reset link	FR-AUT-02, FR-AUT-04
`TOKEN_EXPIRED`, `TOKEN_USED`	410	Link expired or already used	FR-AUT-02, FR-AUT-04
`WEAK_PASSWORD`	422	Fewer than 8 characters, or no letter or digit	NFR-SEC-02
`READ_ONLY_FIELD`	422	Student tried to change major or GPA	FR-PRF-02
`WEIGHTS_NOT_100`	422	Grade component weights do not total 100	FR-CYC-02
`GRADES_EXIST`	409	Components changed after a grade was submitted	FR-CYC-02
`RUBRIC_LOCKED`	409	Rubric changed after the first evaluation	FR-CYC-03
`SCHEDULE_LOCKED`	409	Cycle schedule changed after the first placement	FR-CYC-01
`REASON_REQUIRED`, `COMMENT_REQUIRED`	422	Mandatory reason or comment missing	FR-VAC-02, FR-VAC-08, FR-REP-05, FR-WFL-04, FR-WFL-05, FR-CER-03
`FILE_TOO_LARGE`	413	CV over 5 MB or attachment over 10 MB	FR-PRF-03, FR-REP-02
`UNSUPPORTED_FILE_TYPE`	415	Content is not an allowed type (checked by content, not extension)	NFR-SEC-04
`FILE_PURGED`	410	Verification document deleted after the retention period	FR-ORG-08
`TOO_MANY_ATTACHMENTS`	409	More than 3 attachments	FR-REP-02
`MAIN_DOCUMENT_REQUIRED`	422	Final report without its main document	FR-REP-02
`SKILLS_NOT_CONFIRMED`	409	Matching requested before skills are confirmed	FR-MAT-01
`NOT_ELIGIBLE`	409	Major not eligible, GPA below minimum, or training type different from the student's cycle (CR-06); `reason` is `major_not_eligible`, `gpa_below_minimum`, or `training_type_mismatch`. A vacancy not open to the student's university is out of scope (`403 FORBIDDEN`)	FR-VAC-04, FR-MAT-03
`VACANCY_NOT_OPEN`	409	Vacancy not published	FR-VAC-04
`DEADLINE_PASSED`	409	Application after the vacancy deadline	FR-VAC-04
`DUPLICATE_APPLICATION`	409	Second application to the same vacancy	FR-VAC-04
`APPLICATION_LIMIT_REACHED`	409	Already 5 active applications	FR-VAC-04
`ALREADY_PLACED`	409	Student already has a placement in the cycle	FR-VAC-07
`NO_SEATS`	409	Vacancy has no free seat	FR-VAC-07
`SEATS_BELOW_PLACEMENTS`	422	Seats reduced below the confirmed placements	FR-VAC-01
`ACADEMIC_SUPERVISOR_INVALID`	422	Chosen academic supervisor is not an active supervisor of the student's college	FR-VAC-07, FR-VAC-08
`COMPANY_SUPERVISOR_INVALID`	422	Chosen company supervisor is not an active member of the vacancy's company	FR-VAC-01, FR-VAC-07
`SLOT_CLOSED`	409	Submission to a slot closed as missing	FR-REP-07
`NOTHING_TO_SUBMIT`	409	Submit without a draft	FR-REP-03
`REQUIRED_FIELDS_MISSING`	422	Weekly report fields empty	FR-REP-02
`DEADLINE_NOT_PASSED`	409	Close as missing before the deadline	FR-REP-07
`NOT_FLAGGED`	409	Decision on a similarity check that is not flagged	FR-SIM-03
`TRANSFER_TARGET_INVALID`	422	Role transfer to someone who is not an active member of the same college or company with a password	FR-ORG-07
`SUPERVISOR_HAS_STUDENTS`	409	Coordinator role transfer to an academic supervisor who still has active students	FR-ORG-07
`WEEK_NOT_STARTED`	409	Attendance for a future week	FR-ATT-01
`WEEK_LOCKED`	409	Attendance edit after the edit window without an unlock	FR-ATT-02
`ATTENDED_EXCEEDS_SCHEDULED`	422	More attended days than scheduled	FR-ATT-01
`EVALUATION_INCOMPLETE`	422	Unrated criterion or empty comment	FR-EVL-01

Code	HTTP	When	SRS
`ALREADY_REOPENED`	409	Second reopen of an evaluation	FR-EVL-04
`PLACEMENT_NOT_ENDED`	409	Student evaluation of the company before the placement ended	FR-EVL-05
`PRECONDITIONS_UNMET`	409	Grade submit with missing items; includes `missing`	FR-WFL-03
`GRADE_LOCKED`	409	Edit of a Final grade	FR-WFL-05
`GRADE_NOT_FINAL`	409	Certificate for a grade that is not Final	FR-CER-01
`CERTIFICATE_EXISTS`	409	A valid certificate already exists for the placement	FR-CER-01
`CERTIFICATE_REVOKED`	409	Revoking a certificate that is already revoked	FR-CER-03
`FEEDBACK_NOT_OPEN`	409	Evaluation of the company or survey before the training end date and the Final grade	FR-EVL-05, FR-DSH-03
`ALREADY_SUBMITTED`	409	A second evaluation of the company for the same placement	FR-EVL-05
`CYCLE_NOT_CLOSED`	409	Anonymizing an open cycle	NFR-PRV-04
`RETENTION_NOT_OVER`	409	Anonymizing before `closed_at` plus the retention period; includes `eligible_at`	NFR-PRV-04
`ALREADY_ANONYMIZED`	409	Anonymizing again with nobody left, reopening an anonymized cycle, or a certificate for an anonymized student	NFR-PRV-04
`CONFIRM_REQUIRED`	422	Anonymizing without `confirm: true`	NFR-PRV-04
`INTERNAL_ERROR`	500	Unexpected server error; the details are only in the server log	NFR-USA-05

`PATCH /settings` reports each refused key in `errors` with the code `UNKNOWN_SETTING` or `OUT_OF_RANGE` under `422 VALIDATION_ERROR`.

6. Verification

- **Permission matrix (FR-AUT-05):** `backend/tests/permissions/matrix.yaml` has one entry per endpoint with the callers allowed past the role gates (Public, an unverified student, ST, CS, CA, AS, CO, PA). One test calls every endpoint as every caller that is not allowed and expects `401` or `403`; a second test fails if a route registered in FastAPI is missing from the matrix or the matrix lists a route that does not exist. Scope cases (own record, another user's record, another college or company) are tested in the module suites, whose test data hold two universities, three colleges, and two companies.
- **State machines (FR-WFL-01):** one test per allowed transition and at least one refused transition from each state, through the endpoints of Sections 3.6, 3.7, 3.9, 3.12, and 3.13.
- **Contract:** CI exports the OpenAPI document from the code (`backend/scripts/export_openapi.py`) and fails if it differs from the committed `frontend/src/api/openapi.json`, then regenerates the TypeScript types from it and fails if they differ from the committed `schema.d.ts`, so frontend and backend stay in step.
- **Changes:** a new or changed endpoint is added to this document and to the permission matrix in the same pull request.